

# KamSoft IF\_average V 5.0x Quick Operating Guide

rev005

IF Average

Window

KamSoft  
v.5.01

FFT resolution

512

Multiplication of average

100

Dynamic averaging

103900

Y\_axis\_Offset

36

Y\_axis\_Gain

30

Reset ave.

Reset backgr.

Stop/start ave.

Aca. backgr.

Save single

Multiple files

Number of files to save

1000

Delay between rec. [s]

0

Path

c:\

Files name

data

Stop

Multiple save

Background file

Backgr. Save

Backgr. Read

Scale correction [dB]

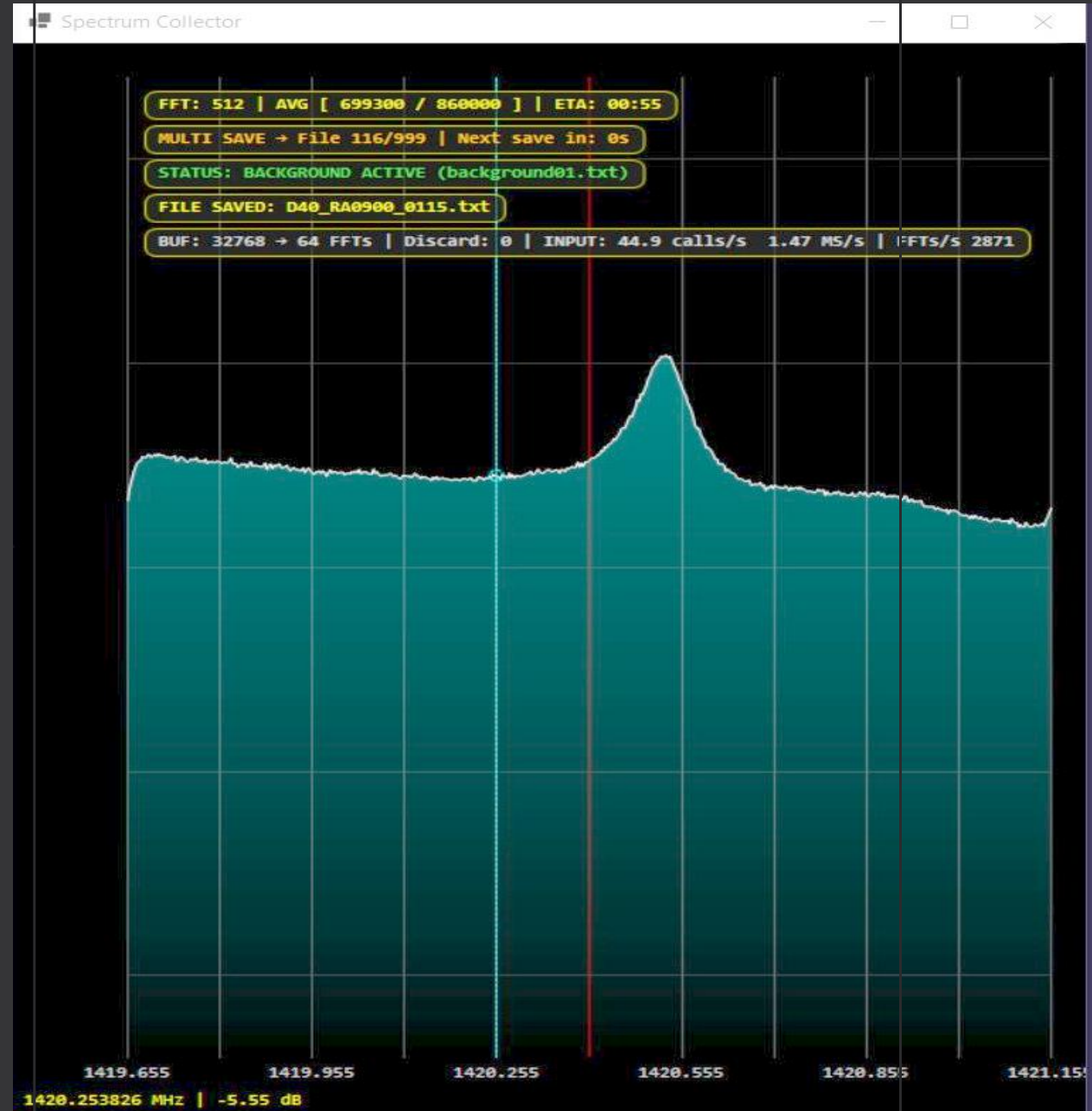
0

☐ Median filter

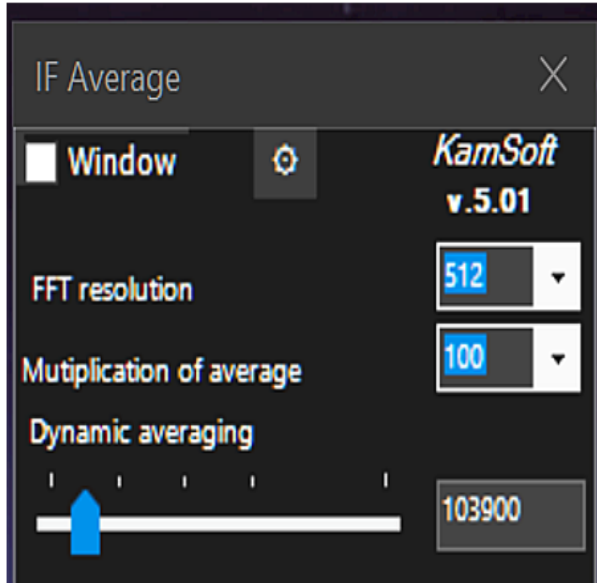
☐ SavitzkyGolay filter

☐ Beep

☒ Stat



# *KamSoft IF\_average V 5.0x Panel*

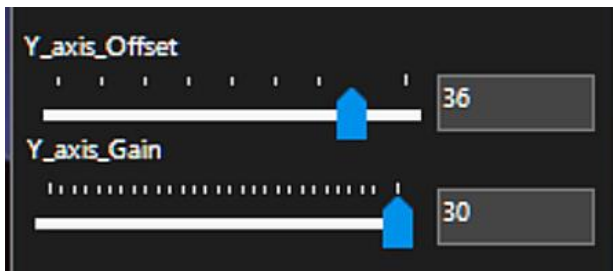


 Open IF\_average Spectral **Window**

**FFT resolution** = # Freq Channels Processed 512 typical

Total number of averages is determined by the  
**Multiplication of average** ( 1 10 100 1000 ) Multiplier \* Dynamic Average

**Dynamic Average** = Total # FFTs averaged per saved file



## Spectral Window Y Axis Range Adjusters

**Y\_axis\_Offset**

**Y\_axis\_Gain**

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## Averaging and Acquisition

**Reset averages**  
**Start/Stop averages**

## Background Correction Functions

**Reset Background**  
**Acquire Background** Background Correction removes  
M curve from Display & Saved data = Use This Option !  
**Save single average**

## Multiple File Save Settings

**Total # files to save** ( can be stopped anytime )  
**Delay between recordings** seconds( Normally **Zero** )  
( this is equivalent to idle time )

**Path** **<< NOTE >>**  
**[ Device and Folder ]** for storing file set >> **Folder *Must Exist* Before Acquiring Data**  
or software will stop after a failed save  
**File name**  
Create a file name which has useful information  
**Stop** acquisition commence **Multiple Save**

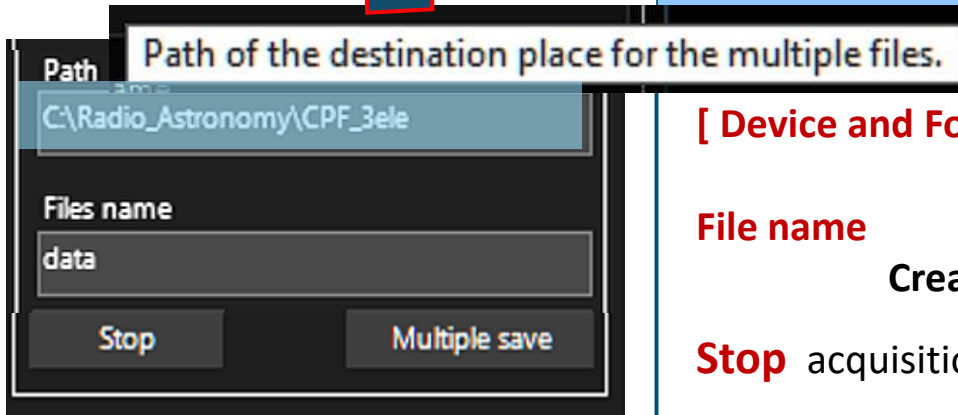
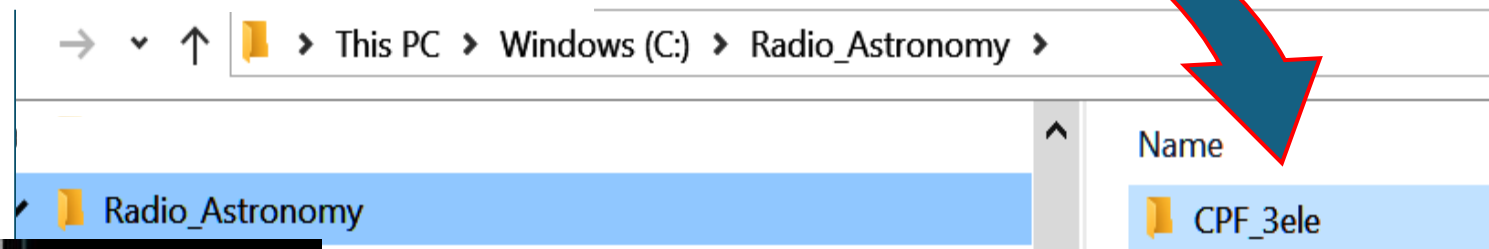
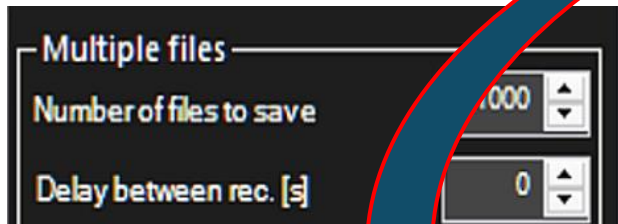
# KamSoft IF\_average V 5.0x Panel

<< NOTE >>

Folder **Must Exist** Before Acquiring/Saving Data

1) Create Folder

2) Click in **Path Window**  
to open Directory  
use IF\_avg  
to Find the Folder



<< NOTE >>

[ Device and Folder ] for storing file set >>

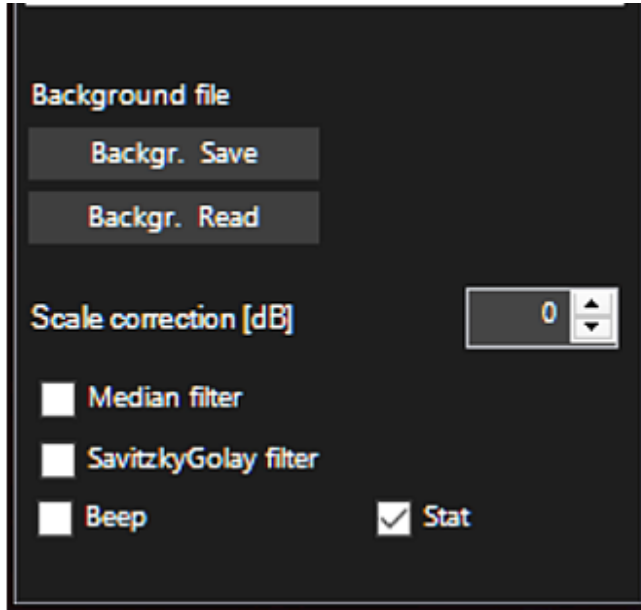
Folder **Must Exist** Before Acquiring Data  
or software will stop after a failed save

**File name**

Create a file name which has useful information

**Stop** acquisition    commence **Multiple Save**

# *KamSoft IF\_average V 5.0x Panel*



## **Background file** operations

**Save** **Mandatory** : begin saved file the name with " **background ..** "

**Read** reuse Saved background correction file

**Scale correction ( dB)** adjust Y\_axis plot labeling

**Median filter** : removes the outlier spectra (single one) from the average.  
it affects saved data.

**S\_G Smoothing filter** : for Spectra Display Window only

**Beep** : sound at each file save operation **Stat** : Display info on Spectra Display Window

# KamSoft IF\_average V 5.0x Panel

IF Average ✕

☐ Window ⚙️ **KamSoft v.5.01**

FFT resolution:

Multiplication of average:

Dynamic averaging:

Y\_axis\_Offset:

Y\_axis\_Gain:

Multiple files

Number of files to save:

Delay between rec. [s]:

Path:

Files name:

Background file

Scale correction [dB]:

☐ Median filter ☐ SavitzkyGolay filter ☐ Beep ☒ Stat

 opens  
plot attributes panel  
for font & color

Plugin Settings ✕

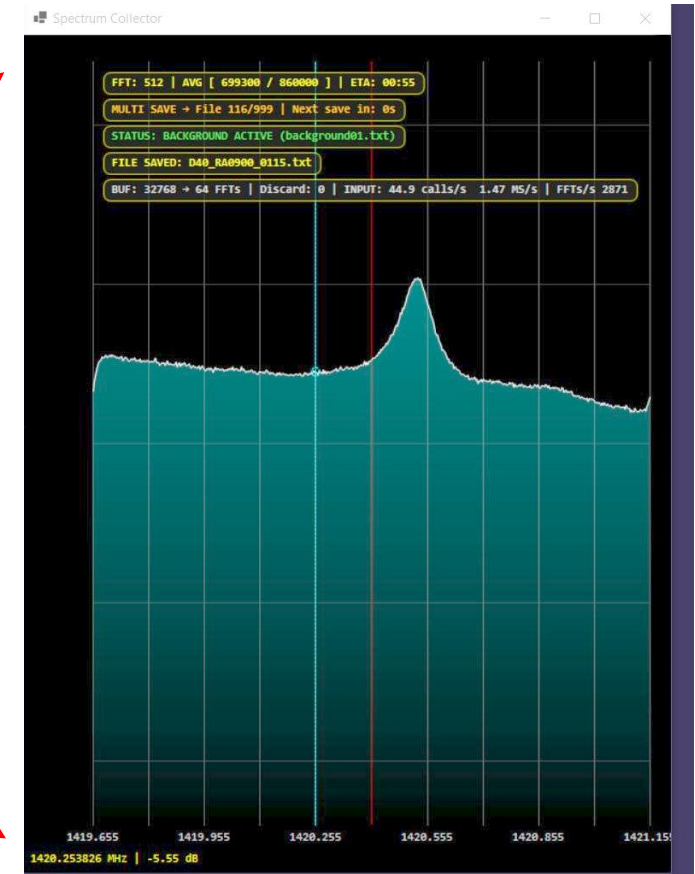
Primary Text Size:

Secondary Text Size:

Grid Brightness:

Text Color:

Graph Color:



## Spectral Window Adjustment XY Cursor Readout Window

Plugin Settings

Primary Text Size:

Secondary Text Size:

Grid Brightness:

Text Color:

Graph Color:

Close

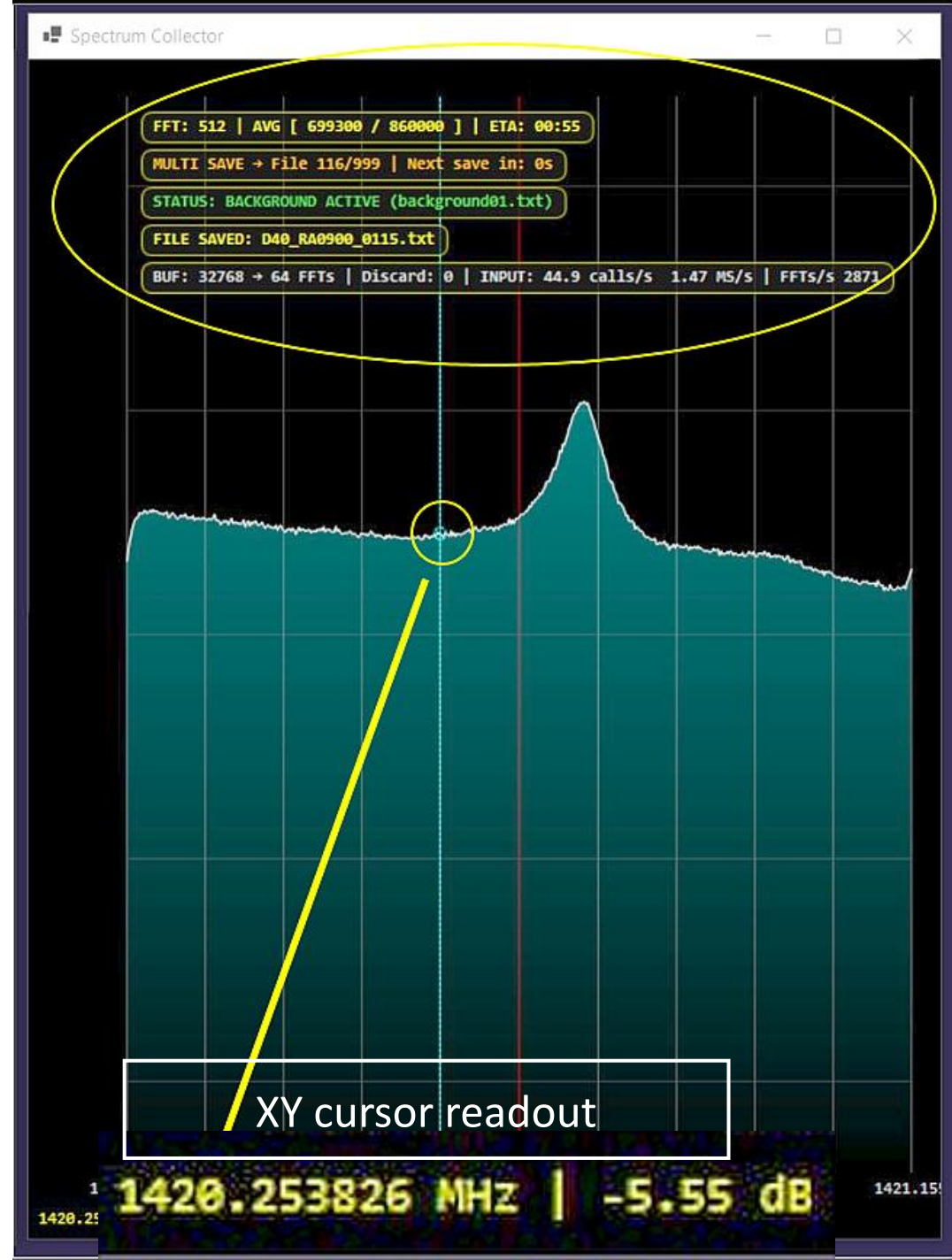
Y\_axis\_Offset

Y\_axis\_Gain

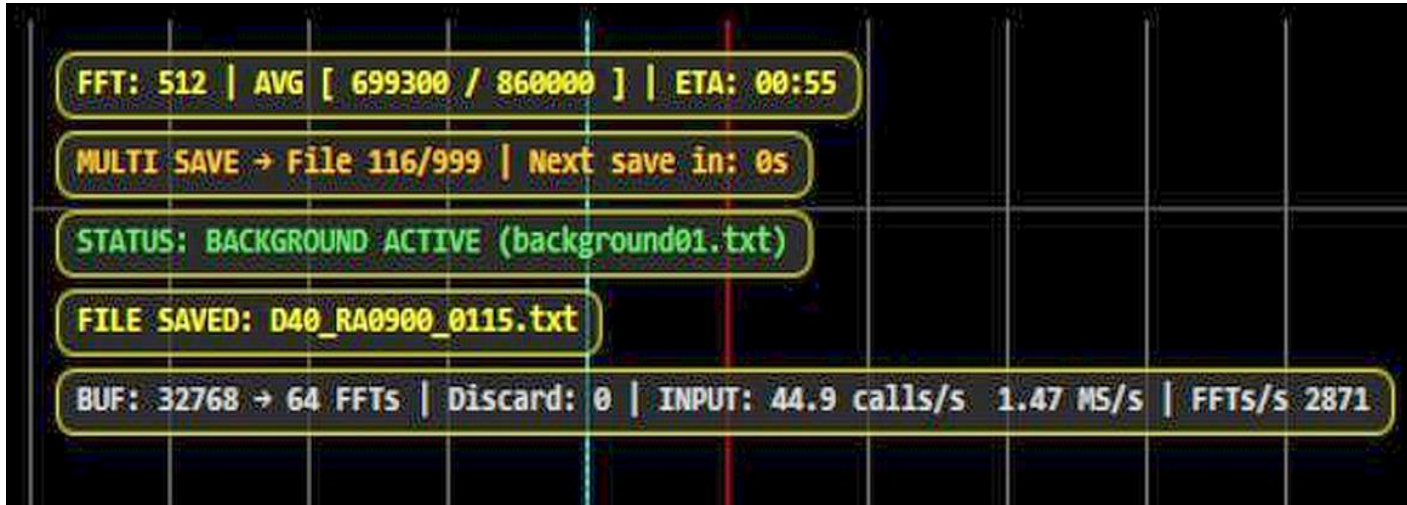
### Spectral Window Y Axis Range Adjusters

Y\_axis\_Offset

Y\_axis\_Gain



# Spectral Display Window **stats panel**



**FFT:** FFT size    **AVG :** [ number acquired / total specified ] | **ETA :** estimated number of seconds to complete AVG

**MULTI SAVE** > nth number file being saved in a multi file sequence | **Next Save :** Delay time seconds until next Avg

**STATUS :** Background Correction active (saved background filename being used )

**FILE SAVED :** name of last file saved ( sequence number auto-appended to primary filename

**Buff**    buffer byte size of data from the device to the plugin through SDR#.

**Discard :** data lost by IFavg plugin.

**Input calls/s** number of calls with the buffer size made per second. Verifies data is being managed correctly.

**MS / Sec** Sample Rate    Msamples/second

**FFTs/s** Average Acquisition Rate FFTs per second ( *useful to calc Dyn Avg size. For 1 deg of sky drift multiply \* 240* )